

Brainstorming & Idea Prioritization Template

Date	14 July 2026
Team ID	SWTID-2026-7049
Project Name	Emotion Detection & Learning Support Engine (AI Learning Assistant)
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing

Each team member listed ideas independently before grouping and judging them.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Parteek Jangir	Detect a student's emotional state directly from their written description of a study problem, using a trained text classifier.	Core ML	1
2	Prateek Singh	Compare two different model architectures (a lightweight BiLSTM and a fine-tuned BERT) side-by-side instead of trusting a single model.	Core ML	1
3	Priyanshu Sharma	Feed the detected emotion into an LLM (Gemini) to generate a supportive, personalized response instead of a generic message.	AI Response	2
4	Khushboo Goyal	Log every interaction to CSV and show an analytics dashboard of emotional trends over time, broken down by field of study.	Analytics	3
5	Aditya Sinha	Let returning students get predictions from previously saved similar examples instead of re-running the models every time.	Nice-to-have	4
6	Parteek Jangir	Let each user bring their own LLM API key so the app isn't limited to one shared free-tier quota.	Nice-to-have	4

Step 2: Idea Prioritization

Each grouped idea was rated on feasibility and importance to select the final direction.

Group No.	Final Idea	Feasibility	Importance	Priority	Selected
1	Dual-model (BiLSTM + BERT) emotion detection with a data-driven trust/decision layer	High	High	1	Yes
2	LLM-generated, emotion-aware supportive responses, with a template fallback when AI is disabled	High	High	2	Yes
3	Session-based interaction logging + a Plotly analytics dashboard	Medium	Medium	3	Yes
4	CSV-based retrieval prediction and bring-your-own API key	Low	Low	4	No

Outcome

The team selected the dual-model emotion detection pipeline (Group 1) and the LLM-based supportive response generator (Group 2) as the core of the solution. Session analytics (Group 3) was included as a supporting feature. CSV-based retrieval prediction and bring-your-own API key (Group 4) were evaluated during development and consciously deferred — the retrieval-prediction idea had no defined matching strategy to build against, and the API-key idea would have required refactoring the LLM clients away from their module-level

singleton pattern this late in the schedule. Both remain visible in the app as clearly labeled, disabled options rather than being silently dropped.